

Evaluation of mRNA Expression of CD244 and Its Adaptor Molecules in CD8⁺ T Cells in Acute Leukemia

Maryam Mohammadi¹, Hossein Asgarian-Omran^{1,2}, Ahmad Najafi¹, Reza Valadan³, Hossein Karami⁴

¹Department of Immunology, School of Medicine, Mazandaran University of Medical Sciences, Sari, Iran; ²Gastrointestinal Cancer Research Center, Mazandaran University of Medical Sciences, Sari, Iran; ³Molecular and Cell-Biology Research Center, Mazandaran University of Medical Sciences, Sari, Iran; ⁴Thalassemia Research Center (TRC), Hemoglobinopathy Institute, Mazandaran University of Medical Sciences, Sari, Mazandaran Iran

Corresponding Author:

Department of Immunology, School of Medicine, Mazandaran University of Medical Sciences, Sari, Iran;
Tel.: (+98-11); Fax: (+98-11); E-mail:

Running title: (**max 50 characters including spaces**)

DECLARATION

(If a section does not apply to your study, please put “Not applicable”)

Acknowledgments

Generative AI and AI-assisted technologies

At IBJ, we recognize that AI and AI-assisted tools are increasingly part of the writing process. Authors may use these tools to improve the clarity, grammar, and overall readability of their manuscripts. However, AI must **not** be used to generate original scientific content, perform data analysis, or provide clinical recommendations. **The responsibility for scientific accuracy, interpretation, and conclusions rests entirely with the authors.** For images, the policy is stricter: AI tools may not be used to alter or manipulate scientific figures, including adding or removing features. Only basic adjustments—such as brightness, contrast, or color balance—are acceptable, and even then, the integrity of the original data must remain intact. Whenever AI tools are used, authors must clearly document this in their manuscript. Please specify: **(1) Which tool was used; (2) For what purpose** (e.g., language editing, grammar correction); **(3) How oversight was maintained.** Declaring the use of AI fosters transparency and builds trust among authors, reviewers, editors, and readers. It also ensures compliance with the terms of use of the relevant AI tool.

Ethical approval

All the experimental procedures in this study were conducted in accordance with the Institutional Animal Care guidelines of and approved by the Research Ethics Committee of, City, Country (ethical code:

Consent to participate

All the (e.g. subjects) voluntarily agree to participate in this research. Written informed consents were provided by(e.g. all the subjects) participated in the study.

Consent for publication

All authors reviewed the results and approved the final version of the manuscript.

Authors' contributions

MM:; HAO:; AN: ([see the link for CRediT Classification](#))

Data availability

All relevant data can be found within the manuscript.

[\(If the relevant data are not within the manuscript, please see the link\)](#)

Competing interests

The authors declare that they have no competing interests. [\(If there is competing interests see the link\)](#)

Funding**No financial support**

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Having financial support

This work was supported by the [grant number(s) xxx].

Supplementary information

The online version contains/does not contain supplementary material.

Evaluation of mRNA Expression of *CD244* and Its Adaptor Molecules in CD8⁺ T Cells in Acute Leukemia

ABSTRACT (not exceed 250 words)

Background: This study investigated the role of the ICR, *CD244*, and its adaptor molecules, in CD8⁺ T cells in acute leukemia.

Methods: Blood samples were obtained from 21 ALL and 6 AML patients and 20 control subjects. Relative gene expression of *CD244*, *SAP*, *EAT-2*, and LncRNA-GSTT1-AS1 were evaluated using qRT-PCR.

Results: Expression of *CD244*, *SAP*, and *EAT-2* were significantly lower in CD8⁺ T cells from ALL patients than those from control subjects. Interestingly, the expression of *SAP* was much lower than that of *CD244*, indicating a lower ratio of *SAP* to *CD244*. Also, *SAP* expression was significantly lower in AML patients compared to the control group. Expression of LncRNA-GSTT1-AS1 showed no significant difference in ALL and AML patients compared to control subjects.

Conclusion: The low *SAP/CD244* expression ratio in CD8⁺ T cells in ALL suggests an inhibitory role for *CD244* in ALL.

Keywords:

(3-5 MESH [<https://www.ncbi.nlm.nih.gov/mesh?Db=mesh&Cmd=ShowDetailView&TermToSearc=>] keywords)

(All gene names should be italicized)

1. INTRODUCTION**2. MATERIALS AND METHODS****3. RESULTS****4. DISCUSSION****5. CONCLUSION**

(In-text references should be ordered numerically and in the square brackets)

(Maximum 50 references)

(Use Vancouver style)

(Output Style for EndNote)

REFERENCES

1. Rose ME, Huerbin MB, Melick J, Marion DW, Palmer AM, Schiding JK, et al. Regulation of interstitial excitatory amino acid concentrations after cortical contusion injury. *Brain Res.* 2002;935(1-2):40-6.
2. Halpern SD, Ubel PA, Caplan AL. Solid-organ transplantation in HIV-infected patients. *N Engl J Med.* 2002 Jul 25;347(4):284-7.
3. Murray PR, Rosenthal KS, Kobayashi GS, Pfaller MA. *Medical microbiology.* 4th ed. St. Louis: Mosby; 2002.
4. Meltzer PS, Kallioniemi A, Trent JM. Chromosome alterations in human solid tumors. In: Vogelstein B, Kinzler KW, editors. *The genetic basis of human cancer.* New York: McGraw-Hill; 2002. p. 93-113.

FIGURES AND TABLES

(Max number of Tables/High-Quality [300 dpi or more] Figures: 8)

- **Table/Figure font: Times New Roman 10**
- **Table margin: Left to Right**